Structural Isomerism

Isomers → Different Compounds which have same molecular formula.
→ They have different physical properties.

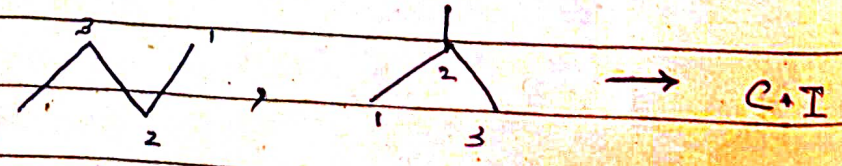
Classification of isomerism.

- Structural (constitutional) iso.
  1. Chain iso.
  2. Position iso.
  3. Functional iso.
  4. Metamerism
  5. Ring chain
  6. Tautomerism (in GOC)
- Stereoisomerism

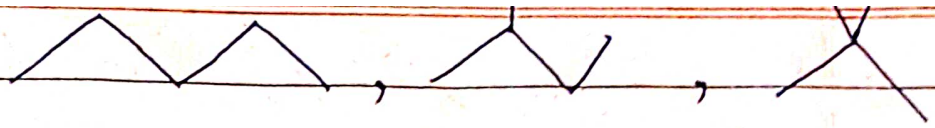
Notable examples:
. dimethyl, trimethyl, tetramethyl benzene = 3 isomers
. use of o,m,p logic on pairs in benzene/ cyclohexane
. always start from DU of compound and make sure the structure satisfies it
. always take care that no carbon has 5 bonds
. use PnC logic like in alcohols & ethers example, carbonyl comp., ester, amines examples
. all possible makes even the impossible possible :))
. for substituents/ multiple bonds start from alkanes and then build upon further possibilities (like in alkenes, hydroxy oic acid examples)

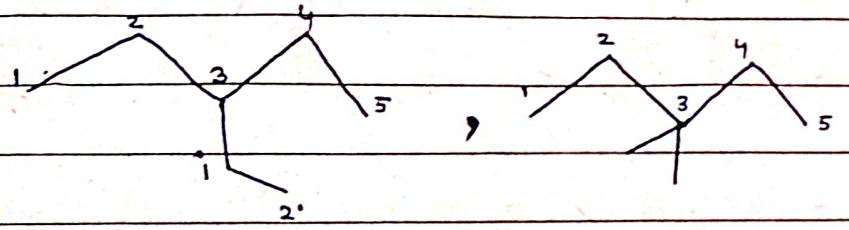
→ Structural Isomers :- Isomers which have different structural formula.
{Different Connectivity between atoms}.

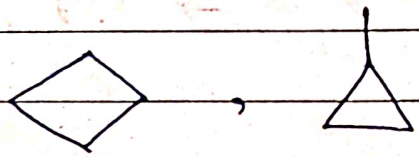
① Chain Isomers → structural isomers which have different size of main chain or chain.

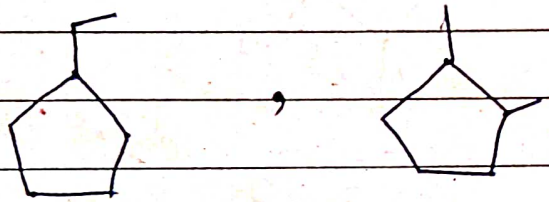
1. C_4H_{10} → 

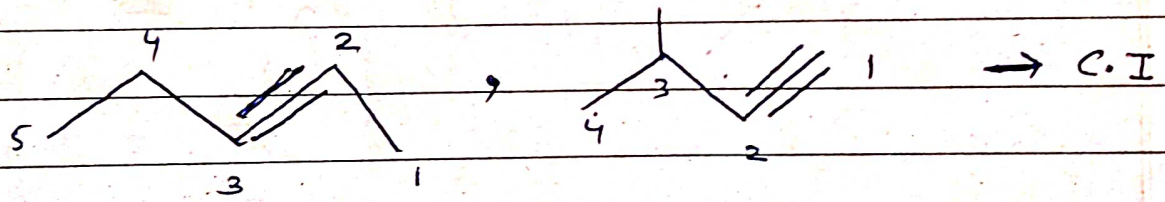
→ C.I

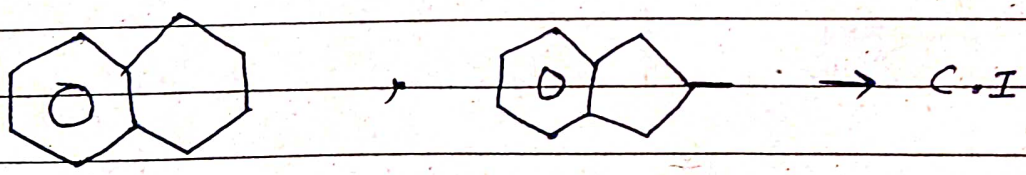
2. C_5H_{12} →  → C.I

3.  → C.I

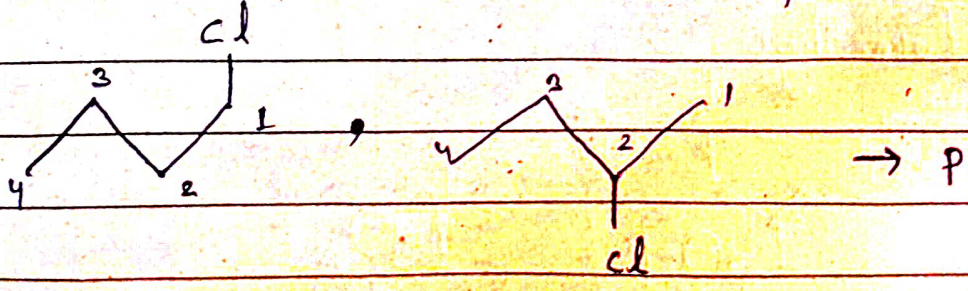
4.  → C.I

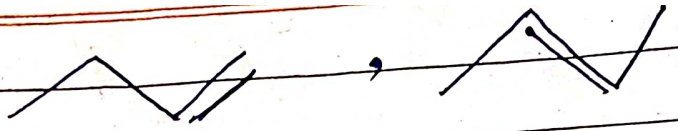
5.  → C.I

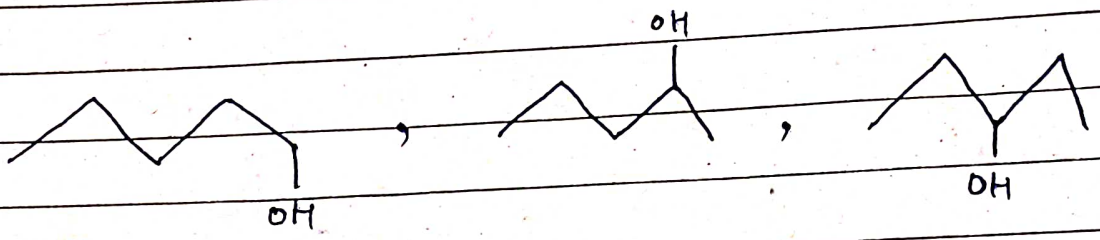
6.  → C.I

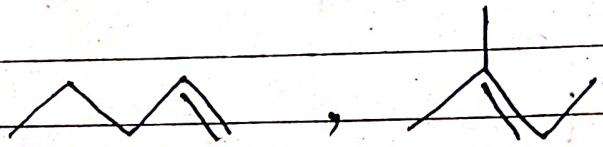
7.  → C.I

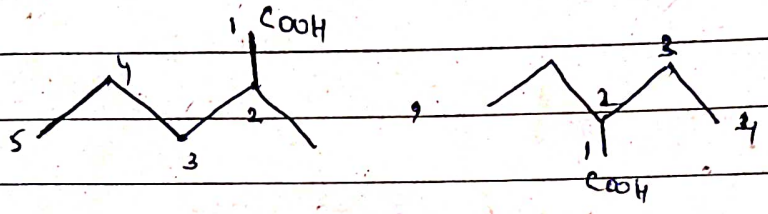
(2) Position Isomers :- Structural isomers which have same size of main chain & side but diff position of F.G, m.b ; substituents.

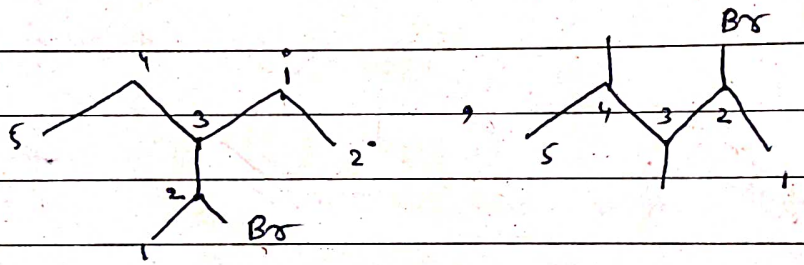
1.  → P.I

2.  → P.I

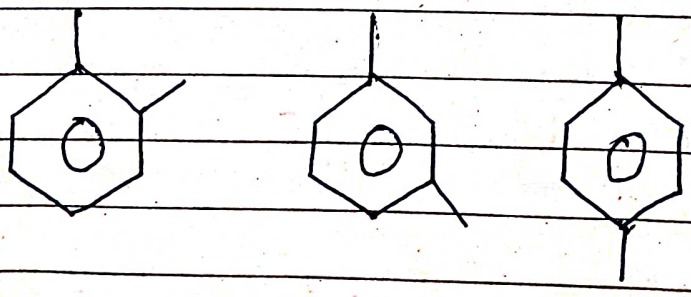
3.  → P.I

4.  → C.I { Not P.I }

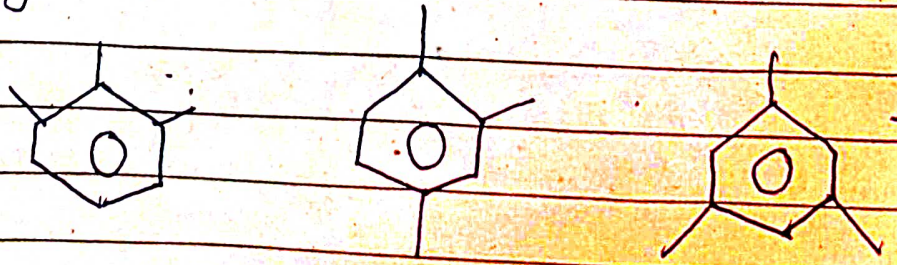
5.  → C.I { Not P.I }

6.  → C.I

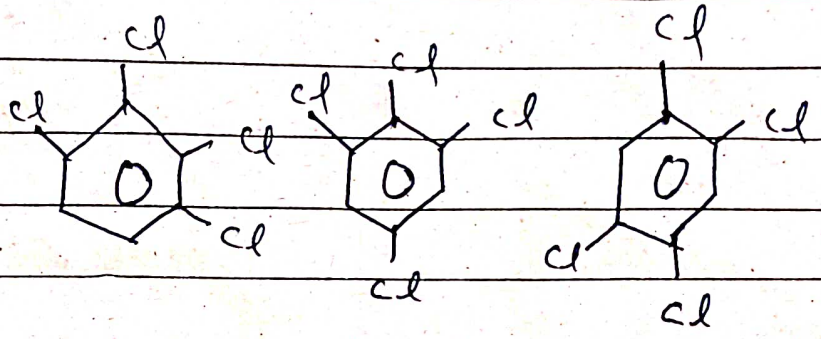
7. Dimethyl benzene

 → P.I

8. Trimethyl benzene

 → P.I

Q. Tetrachlorobenzene

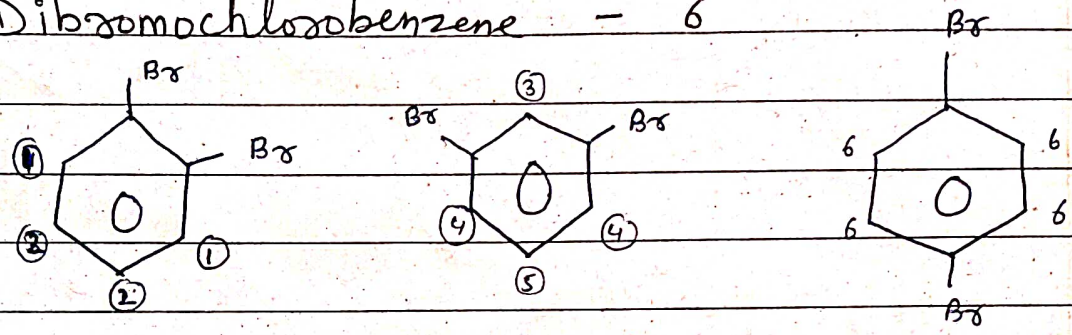


→ P.I

Q.1 Draw all possible position isomers of

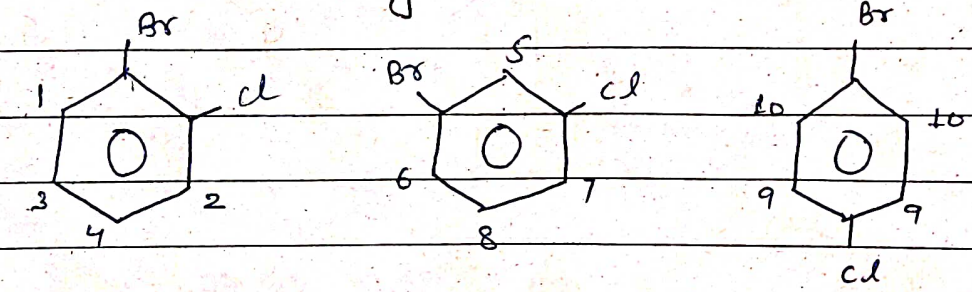
a. Dibromochlorobenzene – 6

Sol²



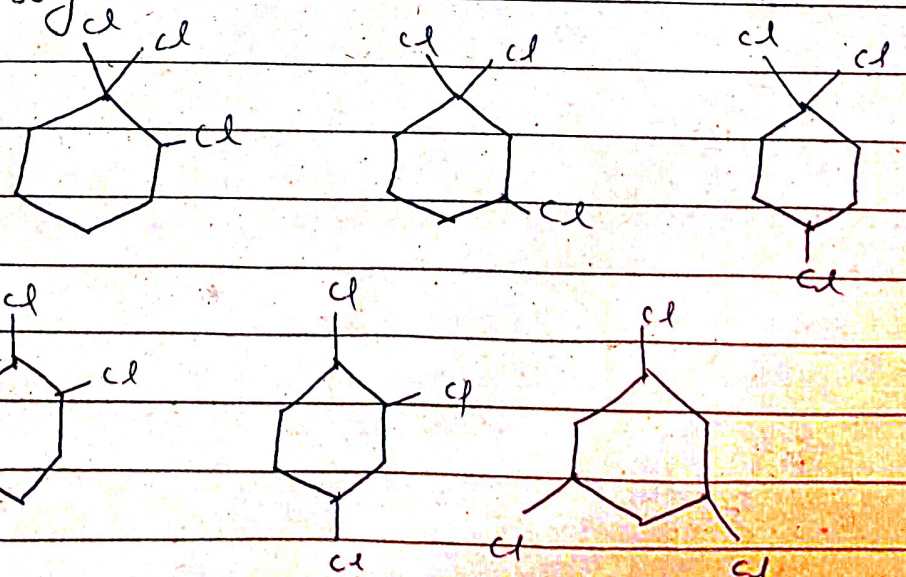
b Bromochloromethylbenzene – 10

Sol²



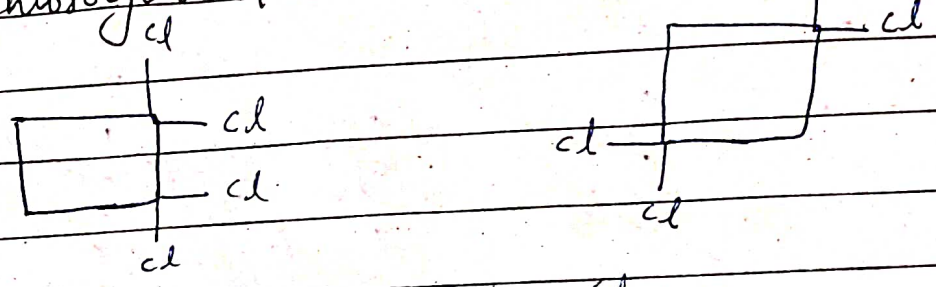
c Trichlorocyclohexane – 6

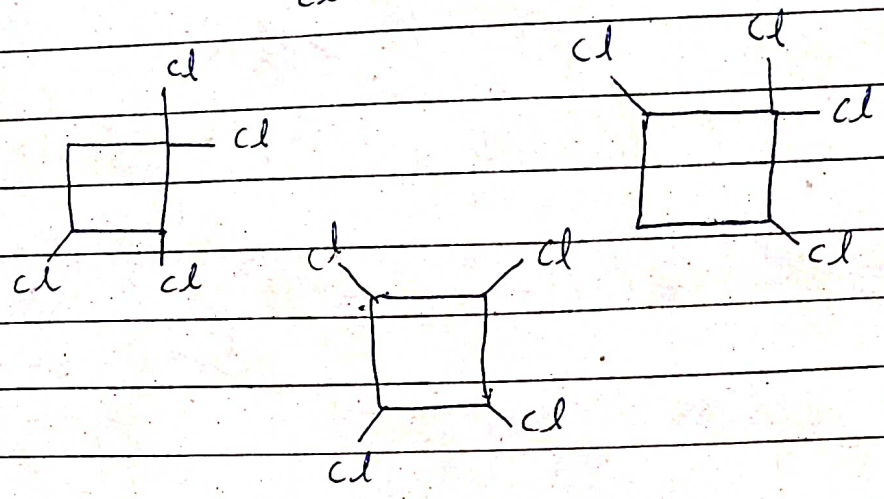
Sol²



d. Tetrachlorocyclobutane – 5

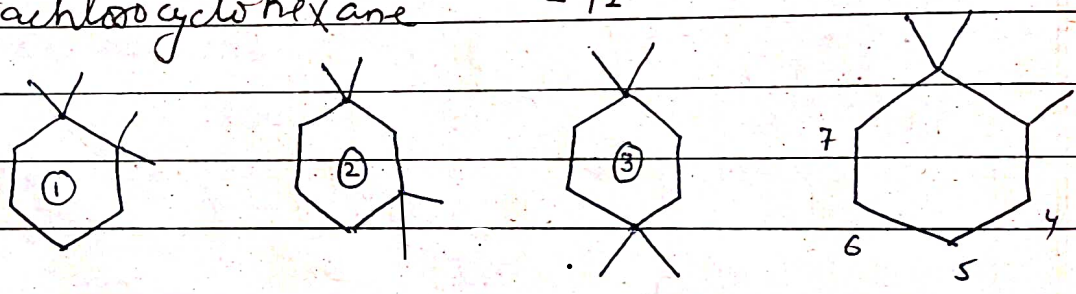
Sol:

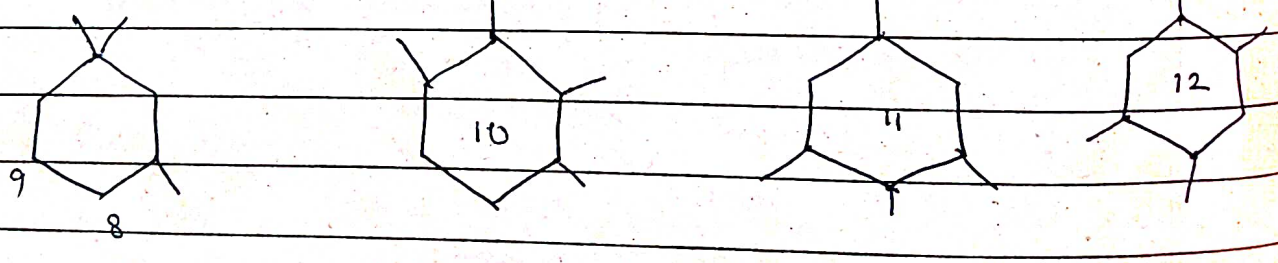




e. Tetrachlorocyclohexane – 12

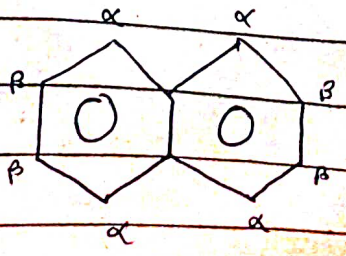
Sol:





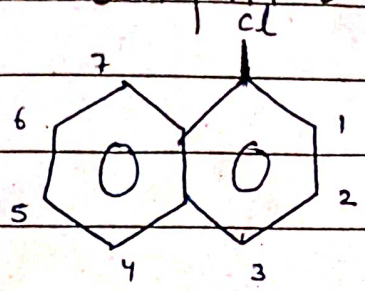
Q2. Draw all possible position isomers of

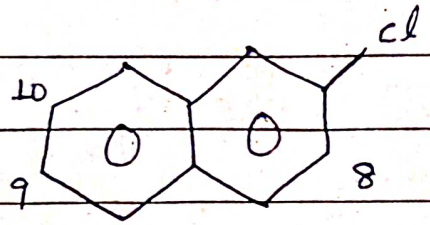
a. Chloronaphthalene – 2



b. Dichloronaphthalene – 10

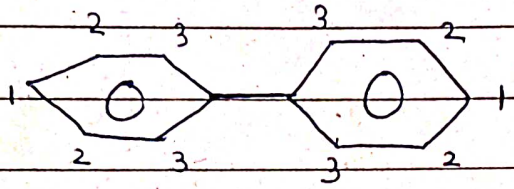
Sol:





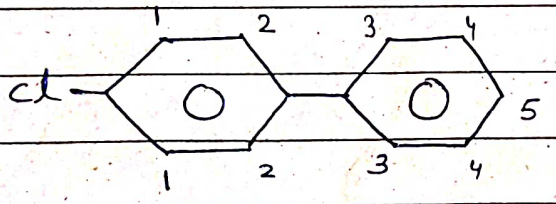
c. Chlorobiphenyl – 3

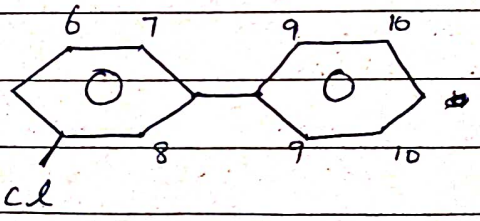
Sol:

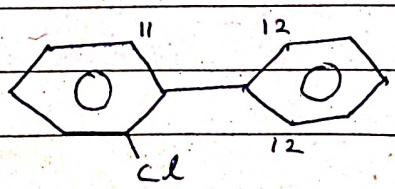


d. Dichlorobiphenyl – 12

Sol:

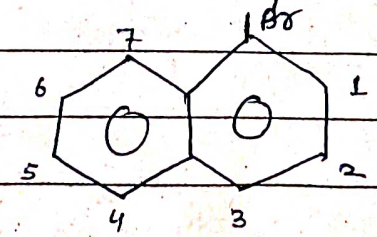


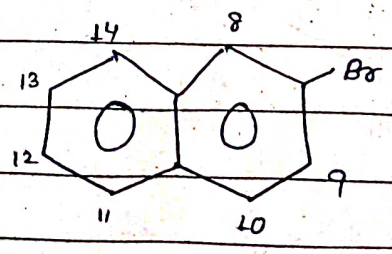




e. Bromo chloronaphthalene – 14

Sol:





Q3. Draw all possible position isomers.

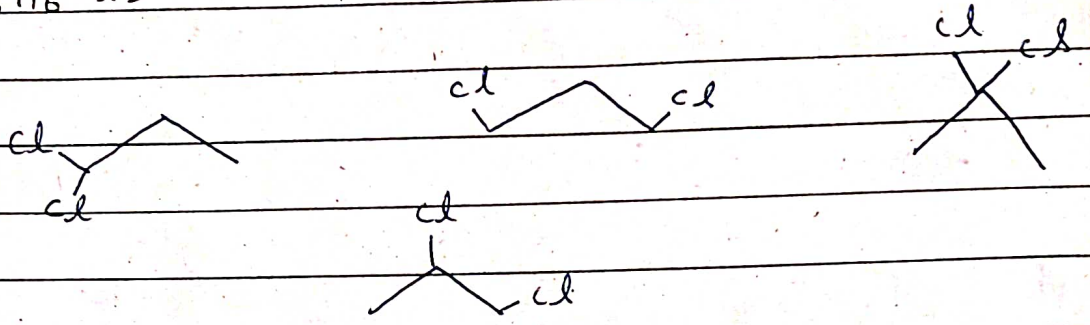
a. C_3H_7Cl – 2

Sol:



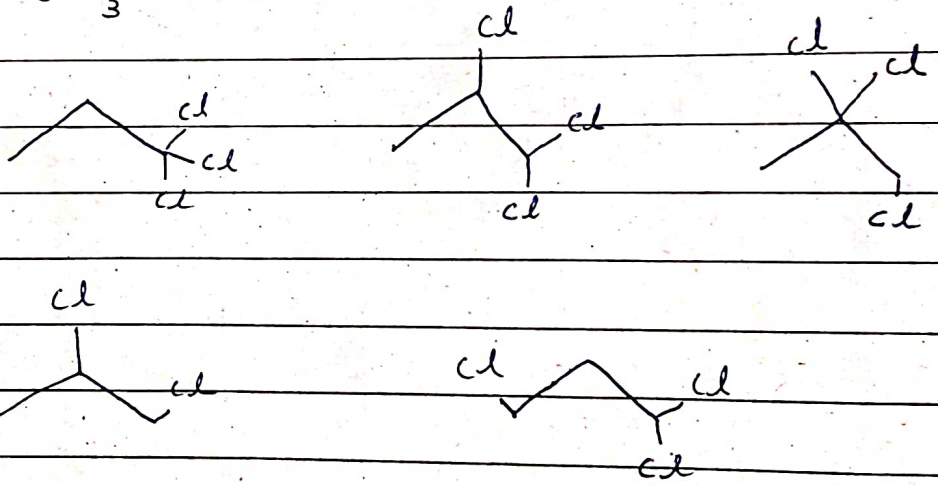
b. $C_3H_6Cl_2$ – 4

Sol:



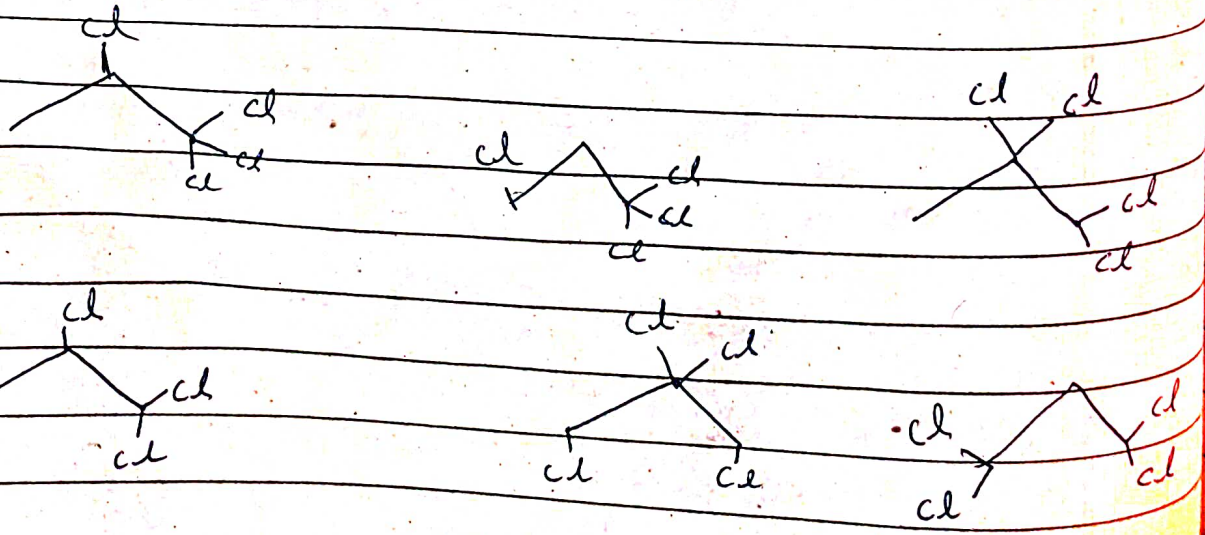
c. $C_3H_5Cl_3$ – 5

Sol:



d. $C_3H_4Cl_4$ – 6

Sol:

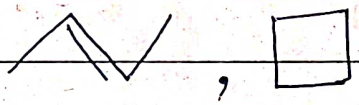


③ Functional isomers :- Compounds which have same molecular formula but have different functional groups.

Note :- Do not Consider position isomerism & size of chain.

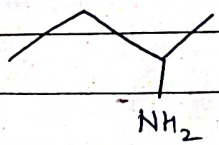
Examples :-

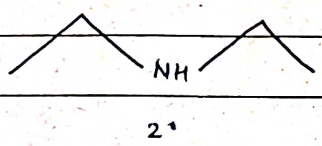
(A) Alkenes, Alkynes & Cycloalkanes.

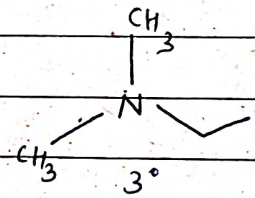
1.  → F.I

2.  , $CH_3-C\equiv C-CH_3$ → F.I

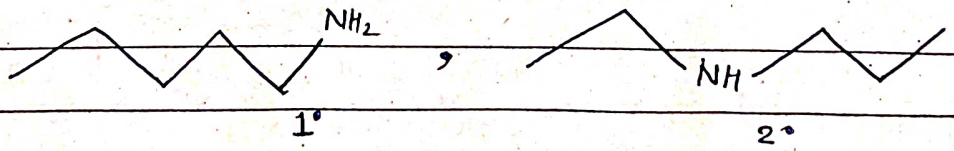
(B) 1°, 2°, 3° amines :-

1. 



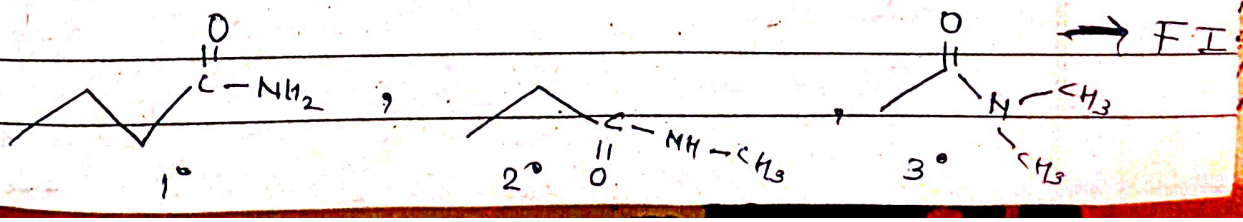


→ F I

2. 

→ F I

(C) 1°, 2°, 3° amides :-

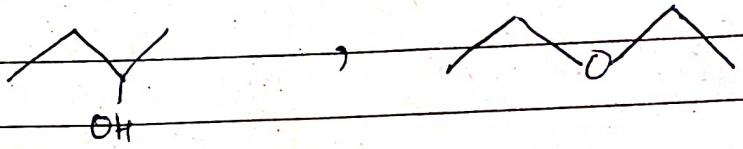
1. 

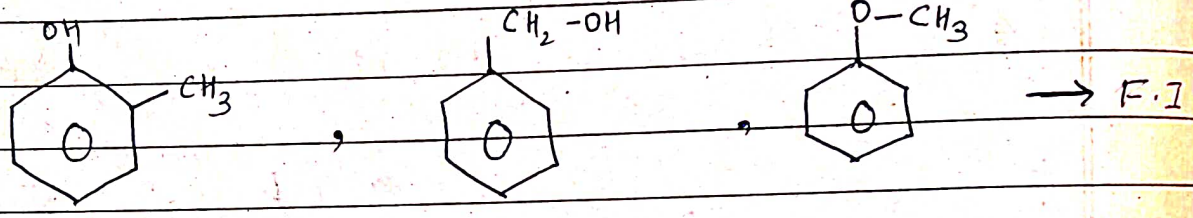
→ F I

(D) Alcohol , Ether & Phenols.

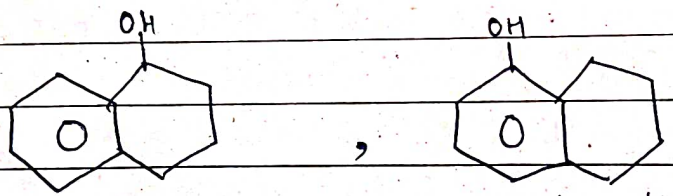
↓ When -OH group is bonded with sp^3 Carbon

↓ When -OH group is bonded with aromatic ring.

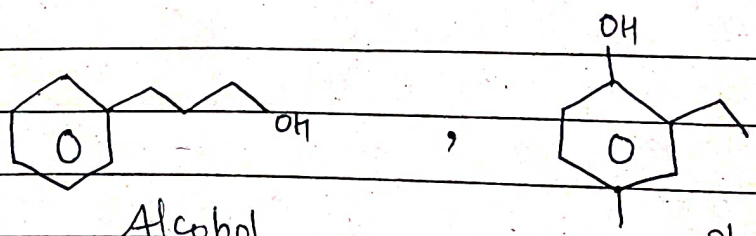
1.  → F.I

2.  → F.I

O-Cresol , Alcohol ,

3.  → F.I

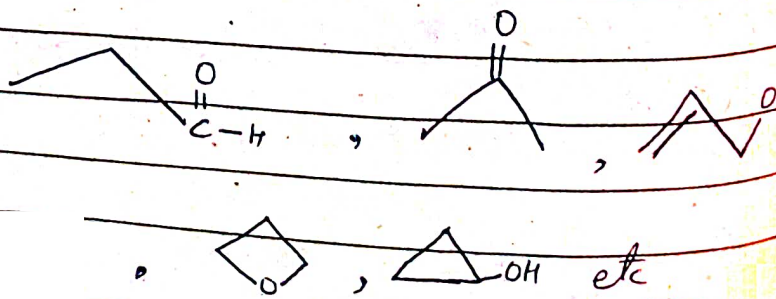
Alcohol , phenolic

4.  → F-I

Alcohol , phenolic

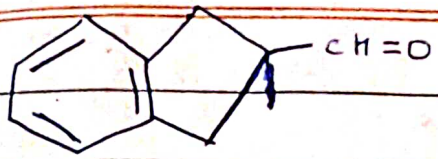
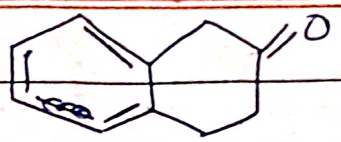
(E) Aldehyde , Ketone , Alkenol , Alkoxyalkene etc.

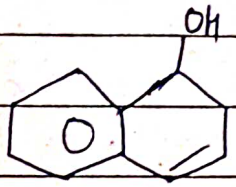
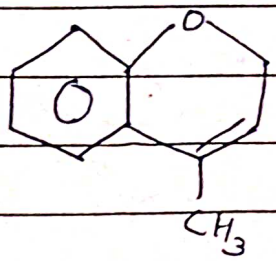
1. C_3H_6O



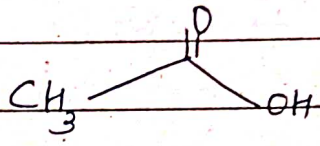
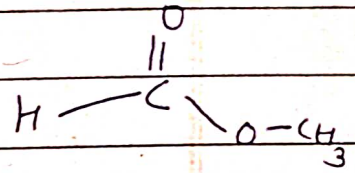
etc

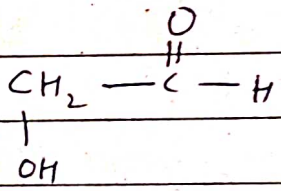
→ F.I

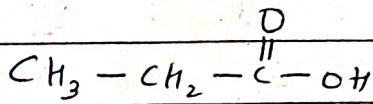
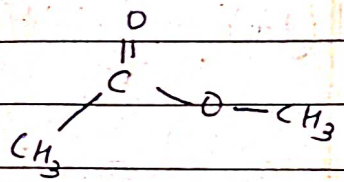
2.  ,  → F.I

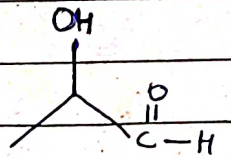
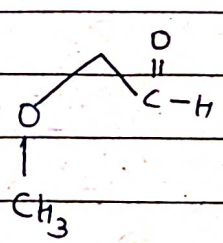
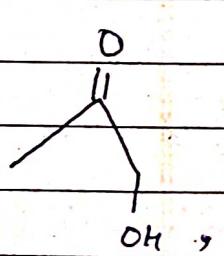
 ,  etc

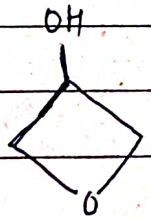
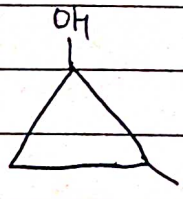
(F) Carboxylic , Esters etc.

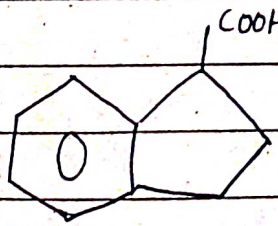
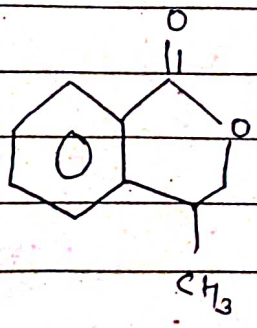
1. $C_2H_4O_2$  , 

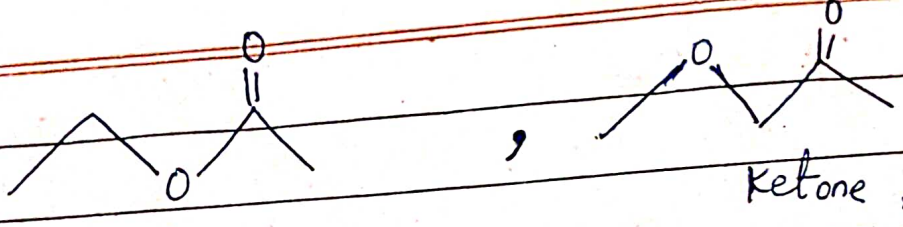
 etc → F.I

2. $C_3H_6O_2$  ,  ,

 ,  ,  ,

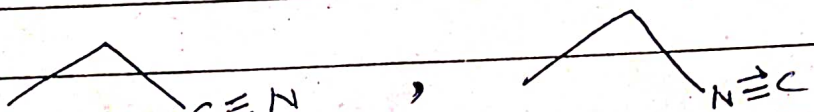
 ,  etc → F.I.

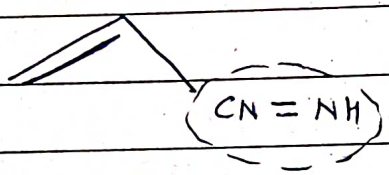
3.   → F.I

4.  → F.I

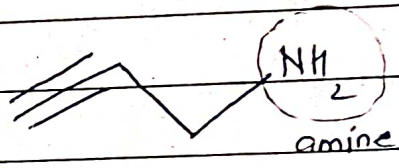
ester , Ketone , ether

(G) Cyanide , isocyanide

1°  → F.I



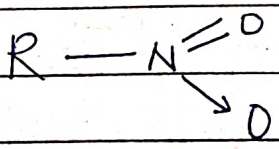
Imine



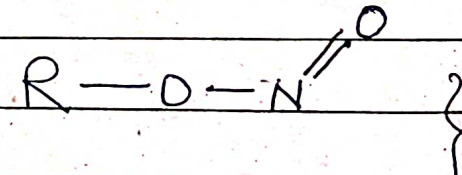
amine

etc

(H) Nitro , Nitrite.



Nitro

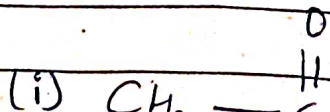
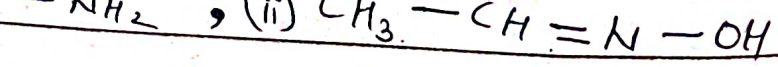


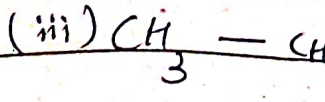
Nitrite

→ F.I

Examples of functional isomers:
a) alkenes, alkynes, cycloalkanes
b) 1°, 2° and 3° amines
c) 1°, 2° and 3° amides
d) alcohol, ether, phenol
e) aldehyde, ketone, alkenol, alkoxy alkenes etc.
f) carboxylic acid, esters etc.
g) cyanide, isocyanide etc.
h) nitro, nitrite
i) amide, oxime

(I) Amide , Oxime.

(i)  , (ii) 

(iii)  etc

→ i, ii → F.I

i, iii → F.I

④ Metamerism :- Isomers which have same polyvalent functional group but have different types of alkyl group.

Note :- Do not Consider position isomerism & size of chain.

⇒ Compounds which Contain Polyvalent functional group Can show metamerism.

Polyvalent functional group → F.Gr which Can divide Carbon chain in more than one part.

* Example of polyvalent F.Gr

1. Ether — $R-O-R$
2. Ester — $R-\overset{O}{\overset{\|}{C}}-O-R$
3. 2° amine — $R-NH-R$
4. 3° amine — $R-\underset{\underset{R}{|}}{N}-R$
5. 2° amide — $R-\overset{O}{\overset{\|}{C}}-NH-R$
6. 3° amide — $R-\overset{O}{\overset{\|}{C}}-N-R$
7. Thioether — $R-S-R$
8. Anhydride — $R-\overset{O}{\overset{\|}{C}}-O-\overset{O}{\overset{\|}{C}}-R$
9. Sulphonic ester — $R-\overset{O}{\overset{\|}{\underset{O}{\overset{\|}{S}}}}-OR$

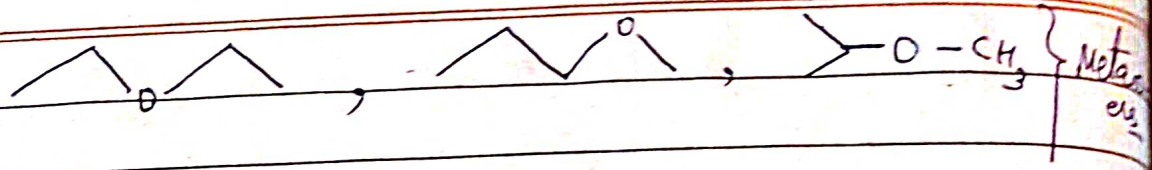
10. Ketones

etc.

{Tauta} F.I > Meta > C.I > P.I Preference Order

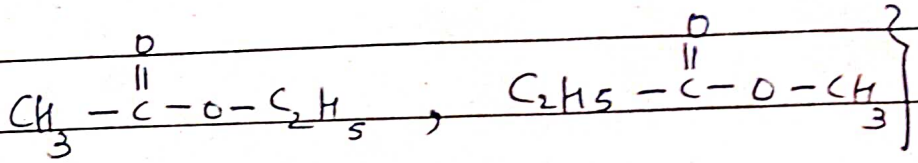
Date :

eg:- 1.



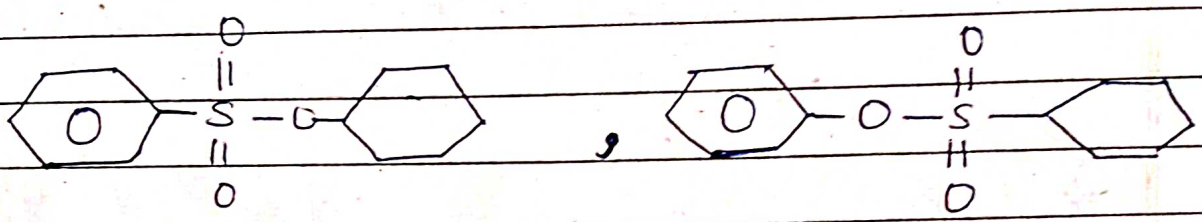
Metamers

2.



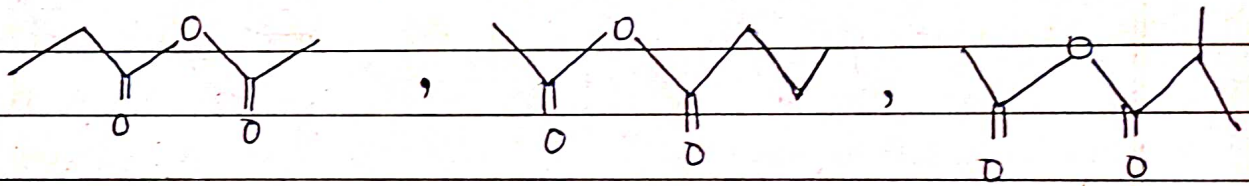
→ Metamers

3.



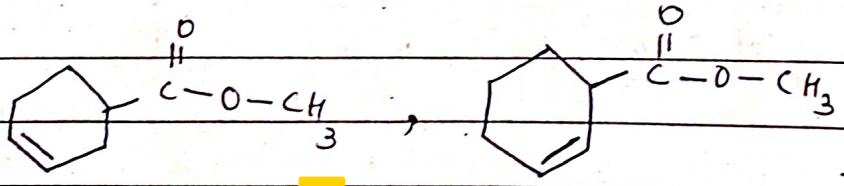
→ metamers.

4.



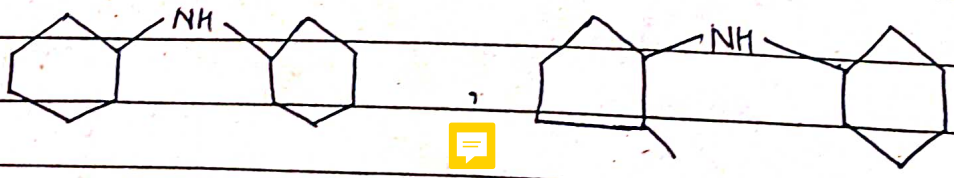
→ metamers.

5.



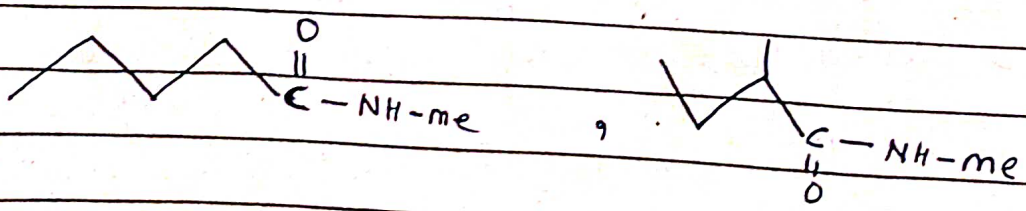
→ metamers.

6.



→ metamer

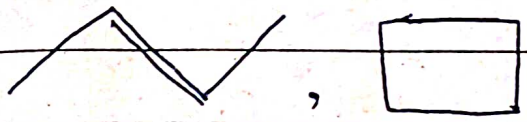
7.

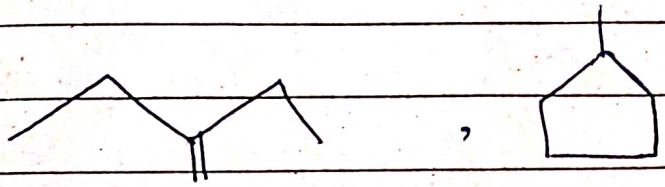


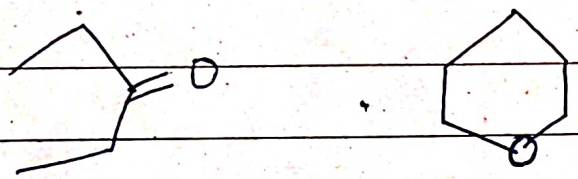
→ metamers.

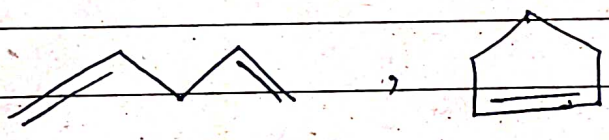
Date :

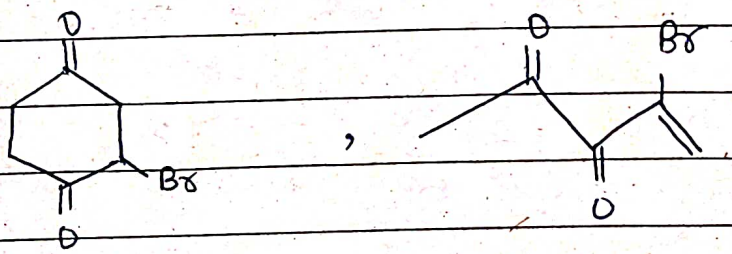
(5) Ring - Chain isomers :-

1.  → F.I & RCI

2.  → F.I & RCI

3.  → F.I & RCI

4.  → RCI.

5.  → RCI & F.I.

Q. Identify all possible structural isomers.

Only ex lid 10 & 11:

Some Important No. of isomers

Alkanes:
C3H8 = 1
C4H10 = 2
C5H12 = 3
C6H14 = 5
C7H16 = 9
C8H18 = 18

Alkyl grps:
C3H7 = 2
C4H9 = 4
C5H11 = 8
C6H13 = 17

STRUCTURAL ISOMERISM

Identify all possible Structural isomers.

① C_3H_8 = 1

② C_4H_{10} = 2

③ C_5H_{12} = 3

④ C_6H_{14} = 5

⑤ C_7H_{16}

⑥ C_8H_{18}

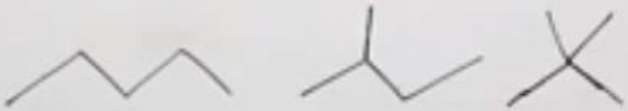
Sol. 1



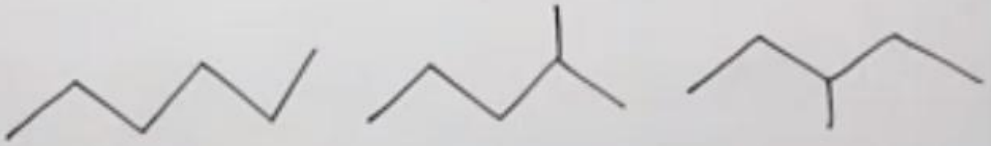
Sol. 2

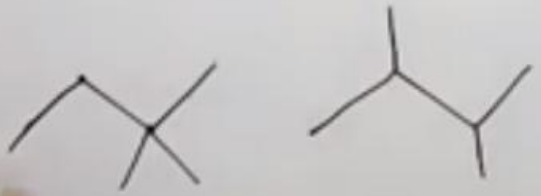


Sol. 3.



Sol. 4





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STRUCTURAL ISOMERISM

Q. Identify all possible Structural isomers.

① C_3H_8 = 1

② C_4H_{10} = 2

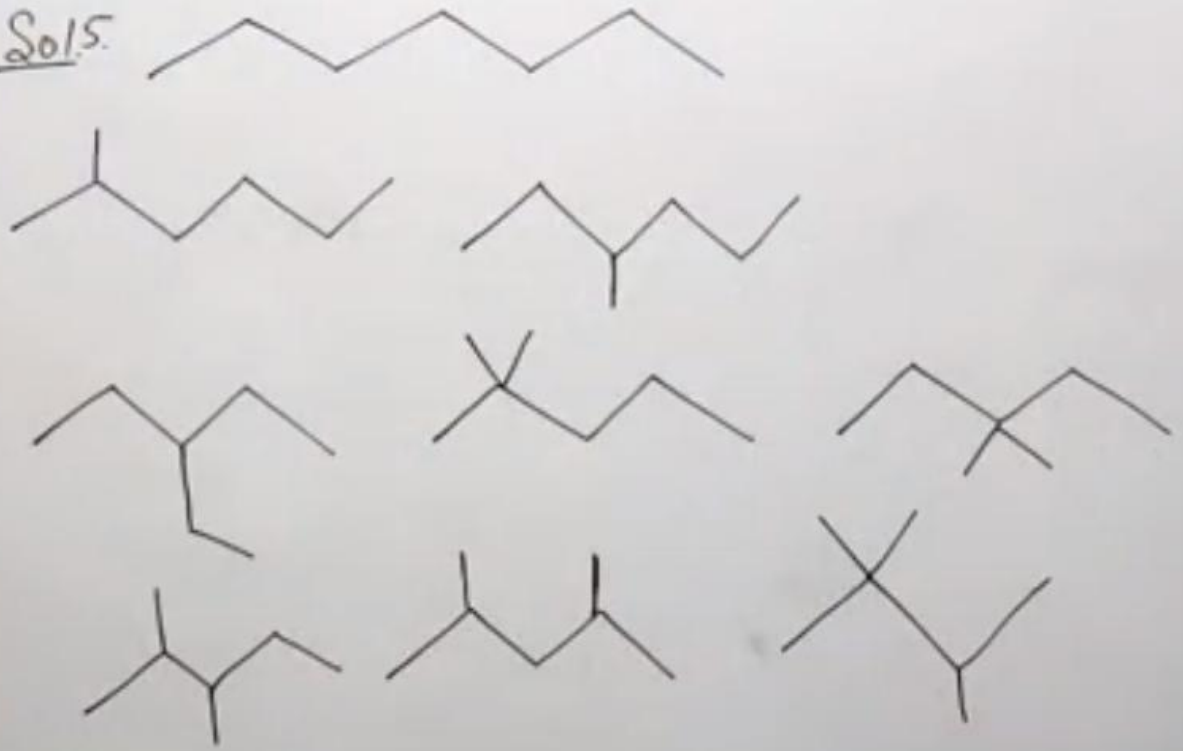
③ C_5H_{12} = 3

④ C_6H_{14} = 5

⑤ C_7H_{16} = 9

⑥ C_8H_{18}

Sol:

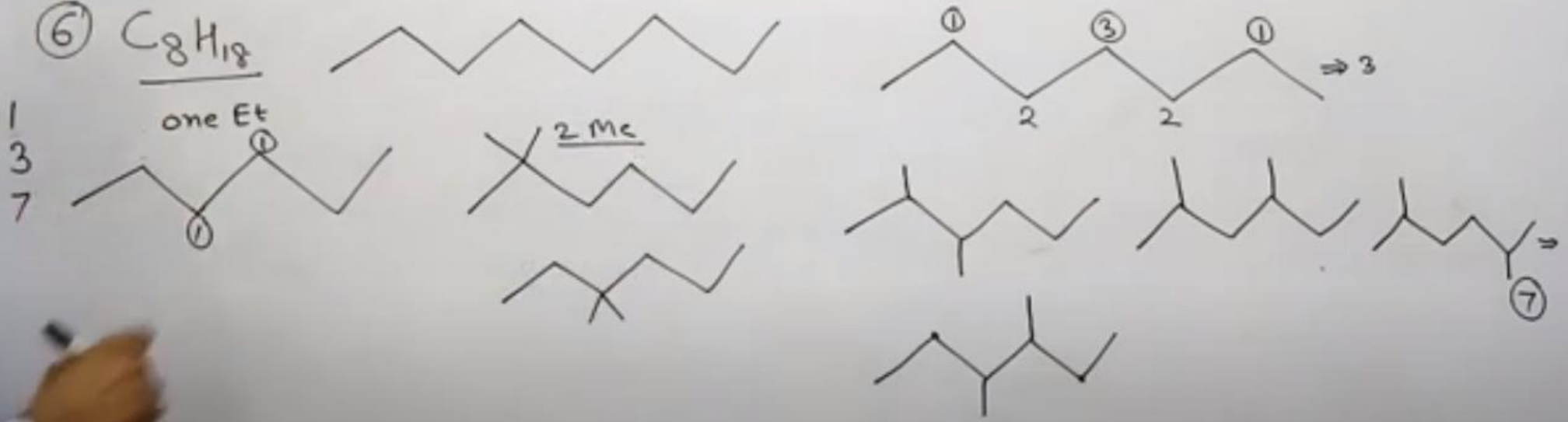


#9|calculation of no of structural isomerism|STRUCTURAL ISOMERISM|JEE MAIN|IIT ADVANCED...

STRUCTURAL ISOMERISM

Q. Identify all possible Structural isomers.

⑥ C_8H_{18}



NEERAJ SAINI (N.S. SIR)

STRUCTURAL ISOMERISM

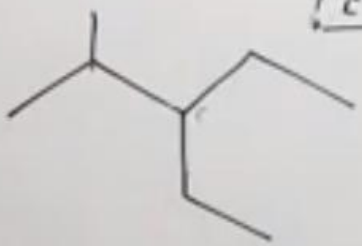
Q. Identify all possible Structural isomers.

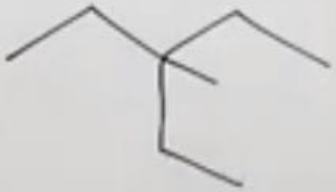
C_3H_7- C-C-C-

C-C(C)-C

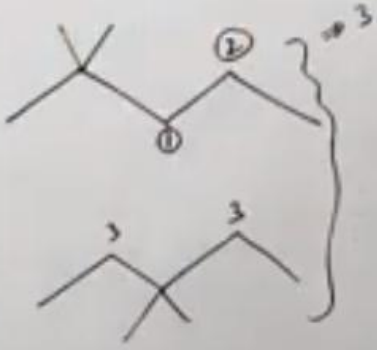
⑥ C_8H_{18}

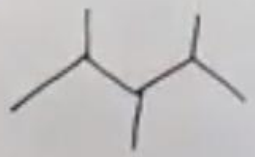
1 Et + 1 Me





3 Me





1
3
7
6
1
18



STRUCTURAL ISOMERISM

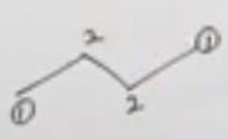
Q2 Identify all possible Structural isomers.

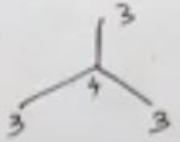
① C_3H_7- ⎫ ⇒2
② C_4H_9- ⎬ alkyl
③ $C_5H_{11}-$ ⎪ ⇒4
④ $C_6H_{13}-$ ⎭

⑤ $C_3H_7-C_9$
⑥ $C_4H_9-C_9$
⑦ $C_5H_{11}-C_9$
⑧ $C_6H_{13}-6$

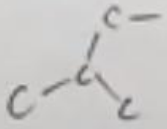
Sol.1 C-C-C- C-C(-)-C

Sol.2



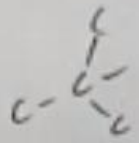


n. C-C-C-C-



Iso.

sec. C-C-C(-)-C



Tert.

Sol.3

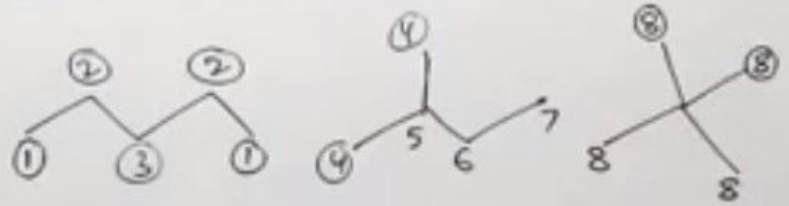
STRUCTURAL ISOMERISM

Q2 Identify all possible Structural isomers.

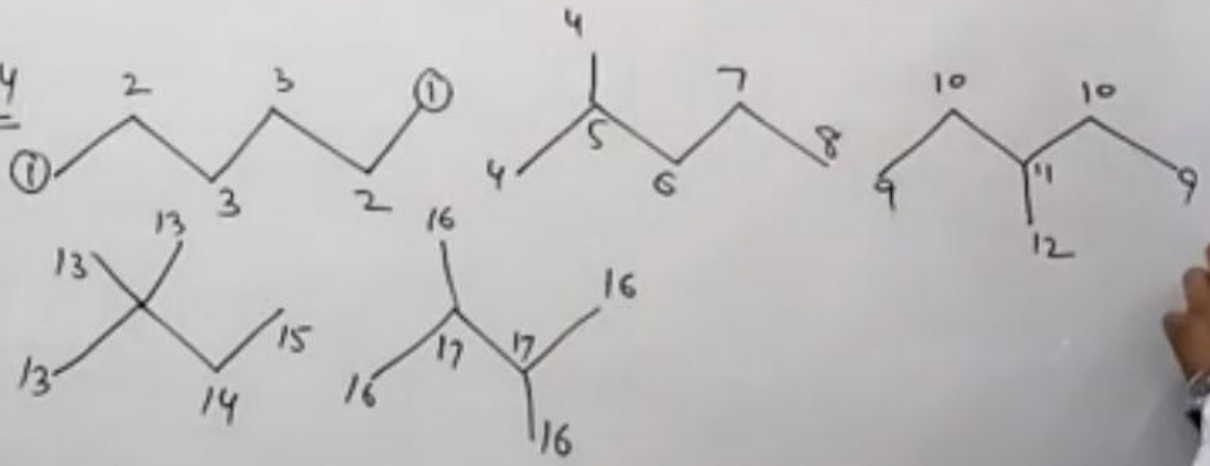
① C_3H_7- ⇒ 2 ✓
② C_4H_9- ⇒ 4 ✓ } alkyl
③ $C_5H_{11}-$ ⇒ 8 ✓
④ $C_6H_{13}-$ ⇒ 17

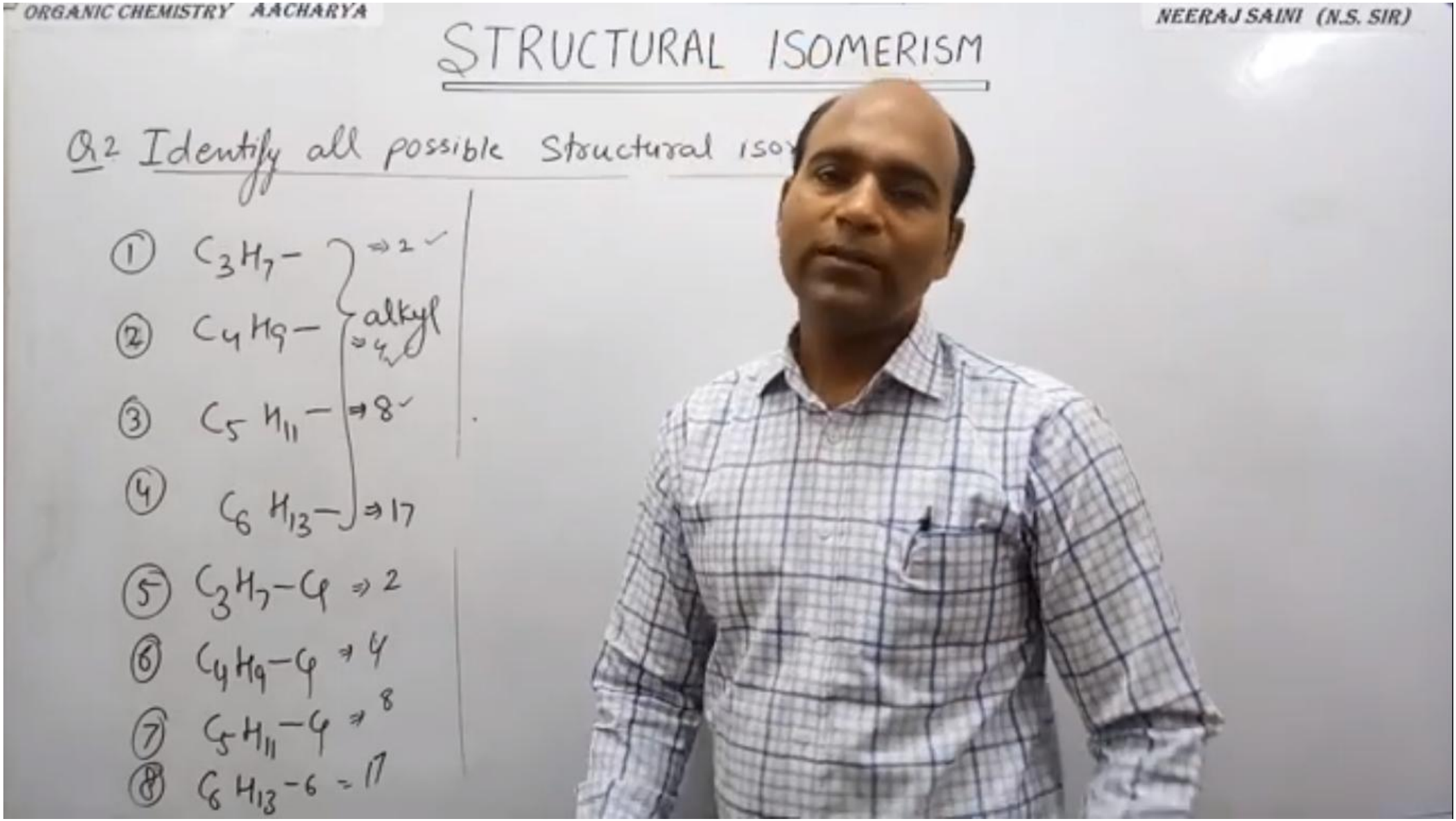
⑤ $C_3H_7-C_6$
⑥ $C_4H_9-C_6$
⑦ $C_5H_{11}-C_6$
⑧ $C_6H_{13}-6$

Sol. 3



Sol. 4




ORGANIC CHEMISTRY AACHARYA
NEERAJ SAINI (N.S. SIR)
STRUCTURAL ISOMERISM
Q2 Identify all possible Structural isom
① C_3H_7- ⇒ 2
② C_4H_9- alkyl ⇒ 4
③ $C_5H_{11}-$ ⇒ 8
④ $C_6H_{13}-$ ⇒ 17
⑤ $C_3H_7-C_6$ ⇒ 2
⑥ $C_4H_9-C_6$ ⇒ 4
⑦ $C_5H_{11}-C_6$ ⇒ 8
⑧ $C_6H_{13}-C_6$ ⇒ 17

STRUCTURAL ISOMERISM

Q3 Identify all possible Structural isomers.

DU

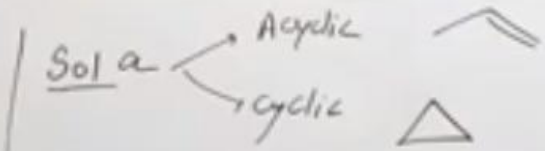
a) C_3H_6 → 2

b) C_4H_8 = 5

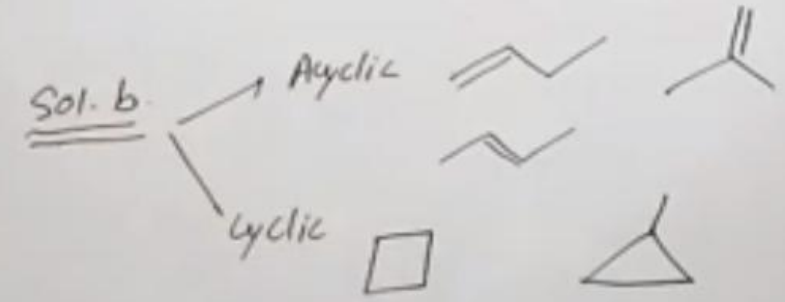
c) C_5H_{10} =

d) C_6H_{12}

Sol a



Sol. b





NEERAJ SAINI (N.S. SIR)

STRUCTURAL ISOMERISM

Q3 Identify all possible Structural isomers.

a) C_3H_6 ⇒ 2

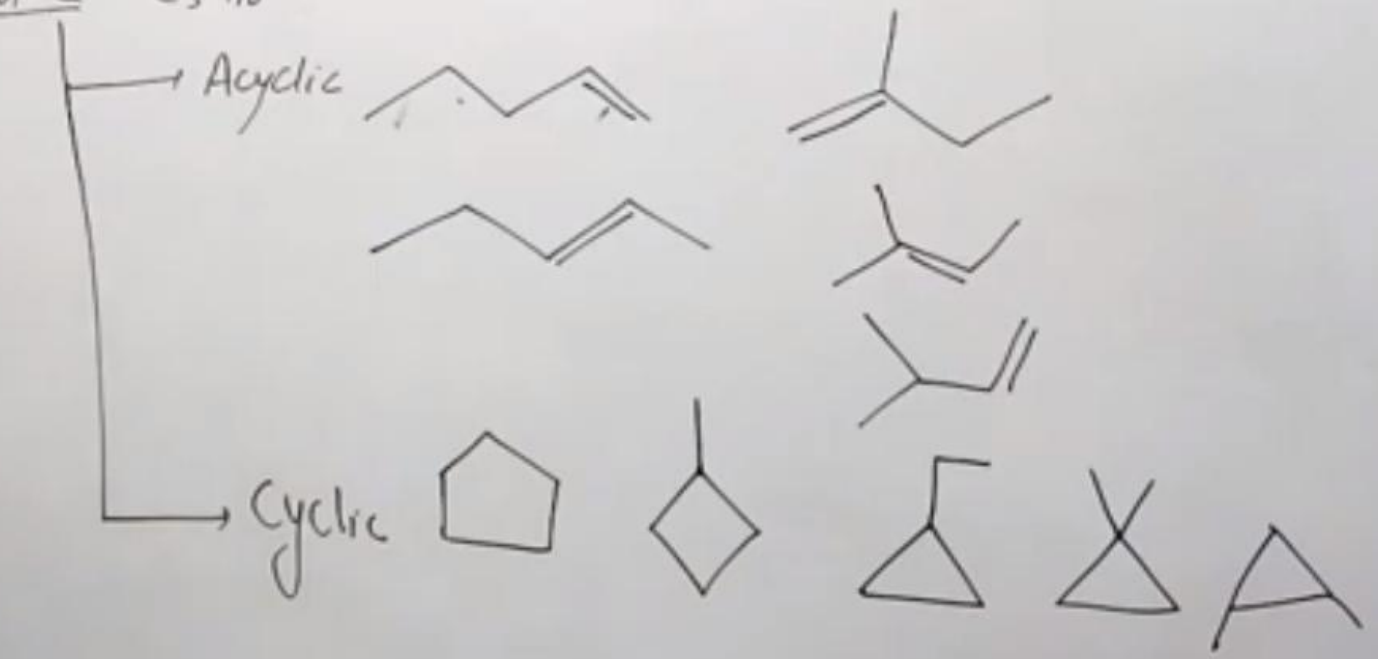
b) C_4H_8 = 5

c) C_5H_{10} = 10

d) C_6H_{12}

DU ①

Sol. c C_5H_{10}



ORGA...ISTRY AACHARYA

NEERAJ SAINI (N.S. SIR)

STRUCTURAL ISOMERISM

...all possible Structural isomers.

...2

...$_8$ =5

...$_{10}$ =10

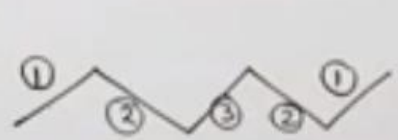
...12

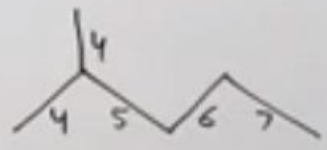
Sold C_6H_{12}

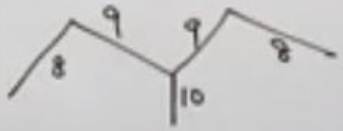
Acyclic=13

Cyclic =

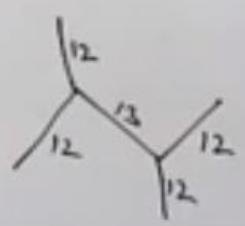
Acyclic











STRUCTURAL ISOMERISM

Identify all possible Structural isomers.

a) $C_3H_6 \Rightarrow 2$

b) $C_4H_8 = 5$

c) $C_5H_{10} = 10$

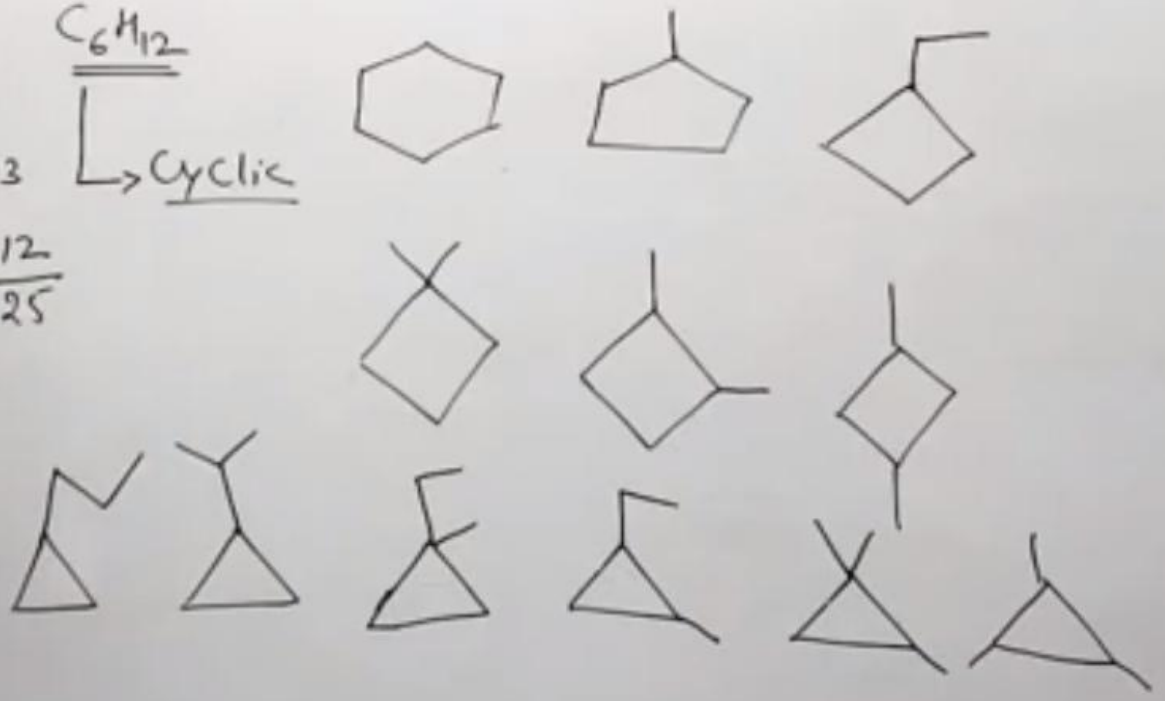
d) $C_6H_{12} = 25$

Sold C_6H_{12}

Acyclic = 13

Cyclic = 12

Cyclic



STRUCTURAL ISOMERISM

Q.4 Identify all possible Structural isomers

DUV 0

a) C_3H_8O → 3

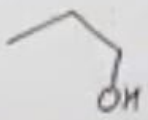
b) $C_4H_{10}O$ -

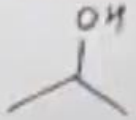
c) $C_5H_{12}O$

d) $C_6H_{14}O$

Sol. a. C_3H_8O

- Alcohol





C_3H_7-OH

- Ether $C_2H_5-O-CH_3$



STRUCTURAL ISOMERISM

Q.4 Identify all possible Structural isomers.

DU = 0

a) C_3H_8O → 3

b) $C_4H_{10}O$ → 7

c) $C_5H_{12}O$

d) $C_6H_{14}O$

Sol. b $C_4H_{10}O$

- Alcohol C_4H_9-OH = 4
- ether $C_3H_7-O-CH_3$ = 2, $C_2H_5-O-C_2H_5$ = 1 } 3



STRUCTURAL ISOMERISM

...ntify all possible Structural isomers.

...O ⇒ 3

$C_4H_{10}O$ - 7

$C_5H_{12}O$ ⇒ 14

: $H_{14}O$

Sol. c $C_5H_{12}O$

- Alcohol $C_5H_{11}-OH$ ⇒ 8
- Ether
  - $C_4H_9-O-CH_3$ ⇒ 4
  - $C_3H_7-O-C_2H_5$ ⇒ 2

} 6

STRUCTURAL ISOMERISM

Q.4 Identify all possible Structural isomers.

a) C_3H_8O → 3

b) $C_4H_{10}O$ - 7

c) $C_5H_{12}O$ ⇒ 14

d) $C_6H_{14}O$ = 32

Sol. d $C_6H_{14}O$

→ Alcohol $C_6H_{13}-OH$ 17

→ Ether $C_5H_{11}-O-CH_3$ → 8

$C_4H_7-O-C_2H_5$ → 4

$C_3H_7-O-C_3H_7$ → 3 (n-o-n, i-o-i, n-o-i)

15

NEERAJ SAINI (N.S. SIR)

STRUCTURAL ISOMERISM

G... possible Structural isomers.

a... 3

only carbonyl comp

only ester

Sol. a) C_4H_8O DU = 1

Aldehyde $C_3H_7-\overset{O}{\overset{\|}{C}}-H$ [2]

Ketone $C_2H_5-\overset{O}{\overset{\|}{C}}-CH_3$ [1]

3

$-\overset{O}{\overset{\|}{C}}-$ carbonyl group.

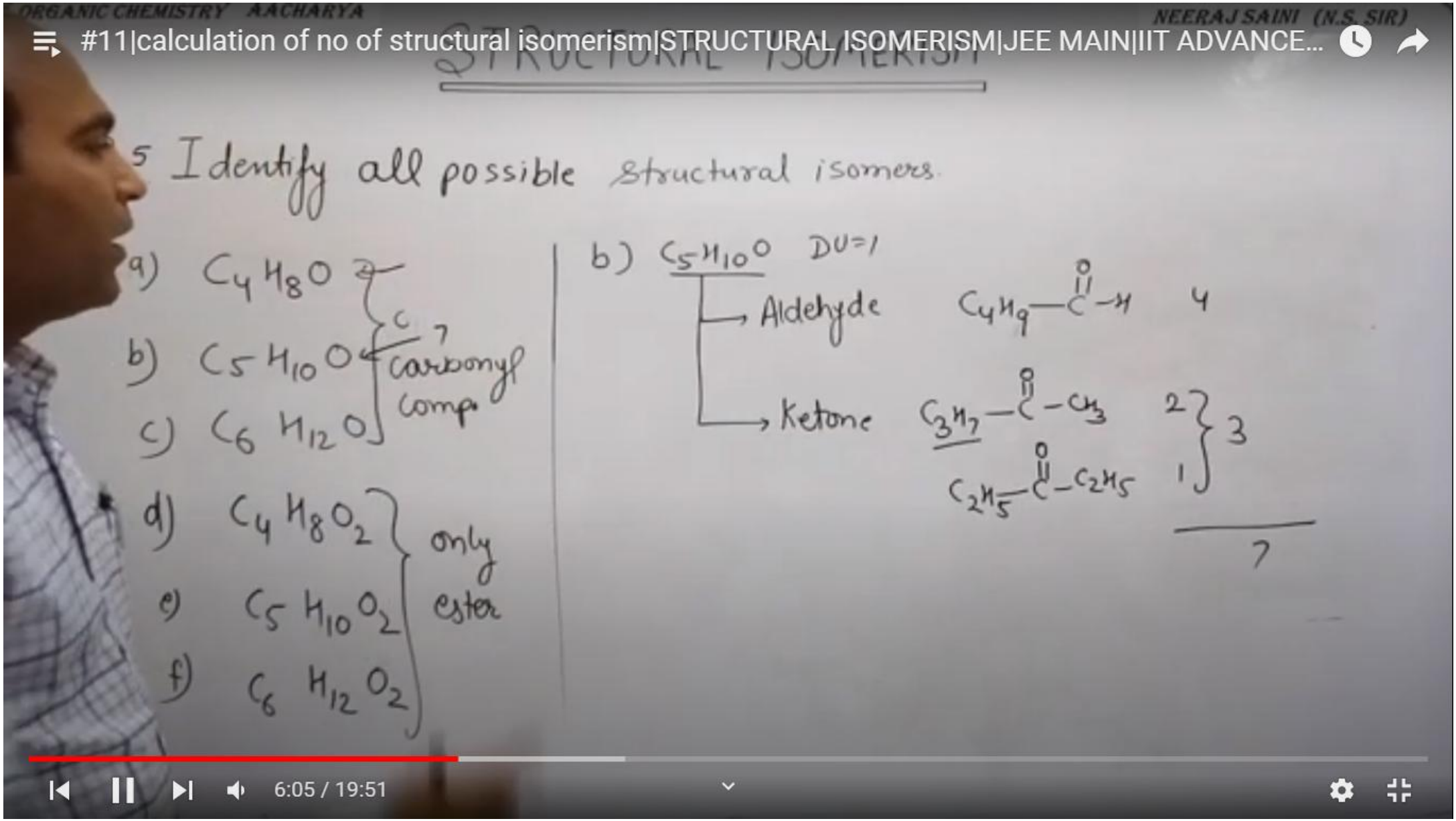
Carbonyl comp.

Aldehyde, Ketone

$C_3H_7- = 2$

$C_4H_9- = 4$

$C_5H_{11}- = 8$


ORGANIC CHEMISTRY AACHARYA
NEERAJ SAINI (N.S. SIR)
#11|calculation of no of structural isomerism|STRUCTURAL ISOMERISM|JEE MAIN|IIT ADVANCE...
STRUCTURAL ISOMERISM
5 Identify all possible Structural isomers.
a) C_4H_8O
b) $C_5H_{10}O$
c) $C_6H_{12}O$
7 carbonyl comp.
d) $C_4H_8O_2$
e) $C_5H_{10}O_2$
f) $C_6H_{12}O_2$
only ester
b) $C_5H_{10}O$ DU=1
Aldehyde
Ketone
C_4H_9-CHO 4
$C_3H_7-CO-CH_3$ 2
$C_2H_5-CO-C_2H_5$ 1
3
7
6:05 / 19:51

STRUCTURAL ISOMERISM

Identify all possible Structural isomers.

a) C_4H_8O → c

b) $C_5H_{10}O$ → c → 7 carbonyl comp.

c) $C_6H_{12}O$ → 14

d) $C_4H_8O_2$

e) $C_5H_{10}O_2$

f) $C_6H_{12}O_2$

} only ester

c) $C_6H_{12}O$ DU=1

→ Aldehyde $C_5H_{11}-\overset{O}{\overset{\|}{C}}-H$ 8

→ Ketone $C_4H_9-\overset{O}{\overset{\|}{C}}-CH_3 = 4$

$C_3H_7-\overset{O}{\overset{\|}{C}}-C_2H_5 = 2$

} 6

14

STRUCTURAL ISOMERISM

Q.5 Identify all possible Structural isomers.

a) C_4H_8O

b) $C_5H_{10}O$ } 7 carbonyl comp.

c) $C_6H_{12}O$ } 14

d) $C_4H_8O_2$ } 4 only ester

e) $C_5H_{10}O_2$ } ester

f) $C_6H_{12}O_2$ }

c) $C_6H_{12}O$ DU=1

→ Aldehyde $C_5H_{11}-\overset{O}{\overset{\|}{C}}-H$ 8

→ Ketone $C_4H_9-\overset{O}{\overset{\|}{C}}-CH_3$ = 4

$C_3H_7-\overset{O}{\overset{\|}{C}}-C_2H_5$ = 2

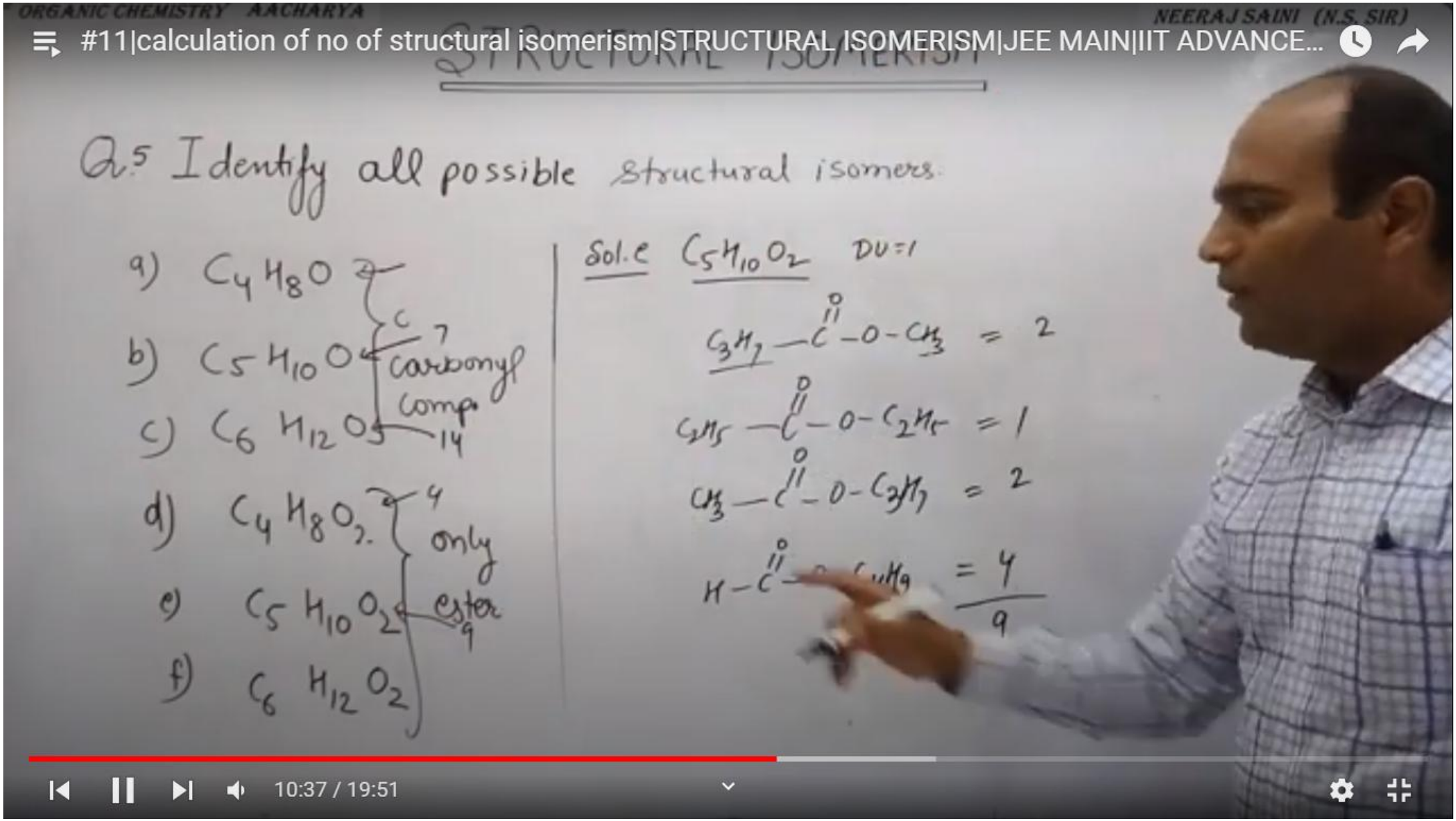
Sol.a $C_4H_8O_2$ DU=1

$C_2H_5-\overset{O}{\overset{\|}{C}}-O-CH_3$ ①

$CH_3-\overset{O}{\overset{\|}{C}}-O-C_2H_5$ ①

$H-\overset{O}{\overset{\|}{C}}-O-C_3H_7$ 2

4


ORGANIC CHEMISTRY AACHARYA
NEERAJ SAINI (N.S. SIR)
#11|calculation of no of structural isomerism|STRUCTURAL ISOMERISM|JEE MAIN|IIT ADVANCE...
STRUCTURAL ISOMERISM
Q.5 Identify all possible Structural isomers.
a) C_4H_8O
b) $C_5H_{10}O$
c) $C_6H_{12}O$
7 carbonyl comp.
d) $C_4H_8O_2$
e) $C_5H_{10}O_2$
f) $C_6H_{12}O_2$
4 only ester
9
Sol.e $C_5H_{10}O_2$ DU=1
$C_3H_7-C(=O)-O-CH_3$ = 2
$C_2H_5-C(=O)-O-C_2H_5$ = 1
$CH_3-C(=O)-O-C_3H_7$ = 2
$H-C(=O)-C_4H_9$ = 4
9
10:37 / 19:51

STRUCTURAL ISOMERISM

Q⁶ Identify all possible Structural isomers.

a) $C_4H_{11}N$ ← 8

b) $C_5H_{13}N$

c) $C_6H_{12}O_3$ Hydroxy carboxylic acid

q) C_4H_6

@ $C_4H_{11}N$ DU=0

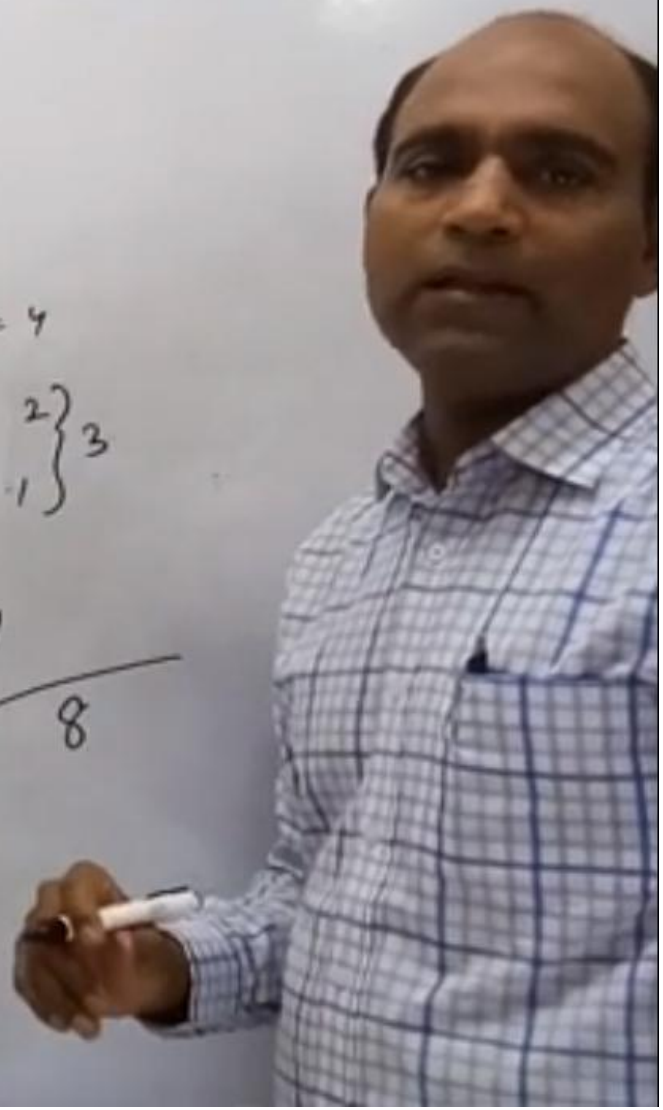
1° $C_4H_9-NH_2$ = 4

2° $C_3H_7-NH-CH_3$ 2

$C_2H_5-NH-C_2H_5$ 1 } 3

3° $CH_3-N(C_2H_5)-CH_3$ 1

8



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STRUCTURAL ISOMERISM

Q⁶ Identify all possible Structural isomers.

a) $C_4H_{11}N$ ← 8

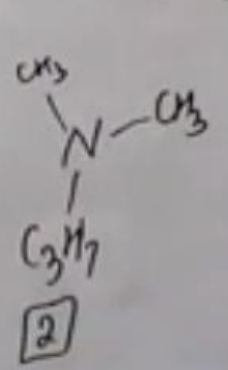
b) $C_5H_{13}N$ ← 17

c) $C_6H_{12}O_3$ Hydroxy carboxylic acid

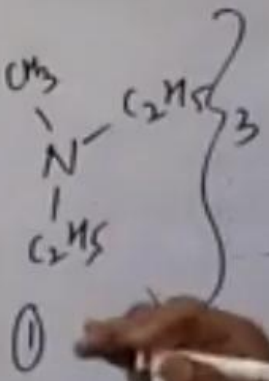
d) C_4H_6

b) $C_5H_{13}N$ DU = 0

- 1° amine $C_5H_{11}-NH_2$ = 8
- 2° amine
  - $C_4H_9-NH-CH_3$ 4
  - $C_3H_7-NH-C_2H_5$ 2
  - } 6
- 3° amine



2



NEERAJ SAINI (N.S. SIR)

STRUCTURAL ISOMERISM

Q⁶ Identify all possible Structural isomers.

a) $C_4H_{11}N$ ← 8

b) $C_5H_{13}N$ ← 17

c) $C_6H_{12}O_3$ ← 29 Hydroxy carboxylic acid

d) C_4H_6

c) $C_6H_{12}O_3$ DU = 1 $-\overset{O}{\overset{\|}{C}}-OH$

